

Section 1: Technical Understanding (40%)

- What was the most surprising discovery about how images are represented?

It's interesting to see the breakdown of how images are processed makes the complex outcome very clear in its steps. Even though the simple R,G,B channels seem like it would limit the ability of the computer to do much with the image's information, combined with analysis algorithms, a lot of useful information can be derived.

- How do the mathematical operations we implemented relate to visual effects?

The mathematical operations simply apply transformations to the pixels in their RGB space. The visual effects are an emergent feature of those transformations. For instance, if we wanted to apply a brightness to an image, we could raise each RGB value together and it would shift toward (255,255,255) which is white. This would brighten the image from what we see.

- Which technique was most challenging to understand and why?

I had a challenge intuitively understanding the histogram data for the image, but I was able to grasp it over time. The visualization of the raw number data of the image does well to explain how the computer is processing the data.

Section 2: Connections and Applications (40%)

- How do today's lab activities connect to the Nano Banana demonstration from class?

Nano banana lets you use generative AI to edit images by prompting a chatbot. The steps from the lab give the AI model the information it needs to begin applying changes. If you prompted, "Remove this person from the photo", nano banana would first approximate the person by using RGB channel analysis, edge detection, etc. and then fill in the image by the same process, writing RGB channels and determining fill shape from information on the surrounding items.

- What real-world applications can you envision for the techniques you learned?

Image editing of course, but also image recognition by using an AI model that is trained on the data ingested by this process.

- How might you combine traditional and AI approaches in a future project?

I'd like to try making a bird/squirrel detector for my backyard bird feeder, and see some historical data on its visitors. Using CV to compare the images with an AI model and also training it to recognize repeat visitors.

Section 3: Personal Reflection (20%)

- What aspect of image processing interests you most for further exploration?

Probably the automation possibility, I've had previous need for quickly cropping images by content, and with some edge detection, I could see it being quite possible. I also need to take pictures of

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device labels a lot for work, having it extract numbers directly into an excel sheet would also be useful.

- How has this lab changed your understanding of digital photography and image editing?

It gave a foundation to form a better intuition for the steps the computer takes when processing an image. Understanding that I could define an editing process through analytical steps rather than a photoshop workflow gives me a new, more powerful tool.

- What questions do you still have about image processing?

I'd like to understand how different file formats handle image information, such as RAW and PNG/JPEG. I think that gets into compression algorithms but it should be a similar process.